

Regression for Regulatory Genomics

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Learning objectives

By the end of this lecture, you will be able to:

- ➊ Apply regression methods to genomic data integration problems
- ➋ Understand fine-mapping and variable selection in regulatory genomics
- ➌ Integrate GWAS and regulatory data using TWAS and colocalization

Roadmap: From biology to methodology

Lecture 12 (Biology)

- Regulatory genomics data types
- ChIP-seq, ATAC-seq, DNA methylation
- Biological interpretation

Today (Methodology)

- Polygenic score model
- eQTLs and chromatin accessibility
- Regression for genomic integration
- Integrative analysis methods

We will discuss multivariate linear regression

Genetic data as an anchor variable

Genetic variants $\rightarrow$ Molecular Phenotype $\rightarrow$ Disease Phenotype

Using 1000 Genomes data via bigsnpr

```
library(bigsnp)  
download_1000G("../data/genotype/")
```

```
data <- snp_attach(.bk.file)  
str(data, max.level=1, strict.width = "cut")
```

```
## List of 3  
## $ genotypes:Reference class 'FBM.code256' [package "bigstatsr"] with 16 fields  
## ..and 26 methods, of which 12 are possibly relevant  
## $ fam      : 'data.frame': 2490 obs. of 6 variables:  
## $ map      : 'data.frame': 1664852 obs. of 6 variables:  
## - attr(*, "class")= chr "bigSNP"
```

Today's lecture

- 1 Polygenic risk prediction ($G \rightarrow \text{phenotype}$)
- 2 Fundamental challenges in PGS (regression)
- 3 Expression Quantitative Trait Loci ($G \rightarrow \text{Expr.}$)
- 4 Co-localization in genomics

Recap: Notations

- X : $n \times p$ genotype matrix of n samples/individuals and p variants
- Y : $n \times 1$ phenotype vector of n samples/individuals

Let's do GWAS to find the disease-associated variants

A variant-by-variant association t-test for a variant j :

$$H_0 : \quad \mathbb{E}[X_{ij} | Y_i = 1] = \mathbb{E}[X_{ij} | Y_i = 0]$$

vs.

$$H_1 : \quad \mathbb{E}[X_{ij} | Y_i = 1] \neq \mathbb{E}[X_{ij} | Y_i = 0]$$

- The average genotype is the same between the case and control under the null.

GWAS statistics over all the variants

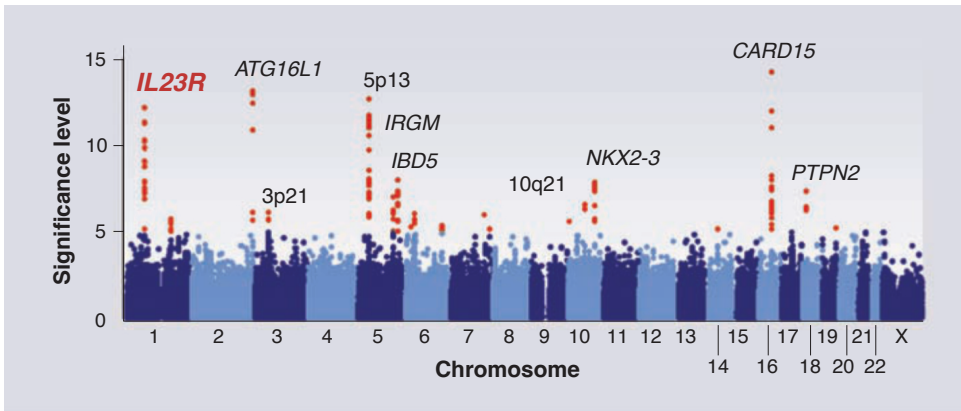
```
## library(matrixTests)
.gwas <- col_t_welch(X[Y == 0, ], X[Y == 1, ])
```

- For each pair of columns in the two matrices, the function performs a t-test and summarizes all the results into a list of vectors.

```
names(.gwas)
```

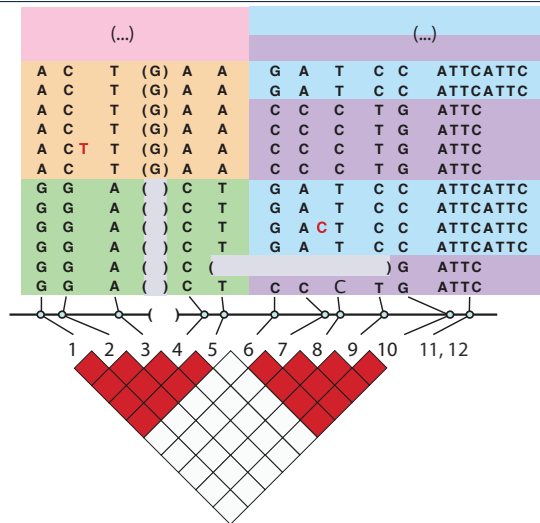
```
## [1] "obs.x"      "obs.y"      "obs.tot"    "mean.x"     "mean.y"
## [6] "mean.diff"  "var.x"      "var.y"      "stderr"     "df"
## [11] "statistic"  "pvalue"     "conf.low"   "conf.high"  "mean.null"
## [16] "alternative" "conf.level"
```

Human genetic variation recap



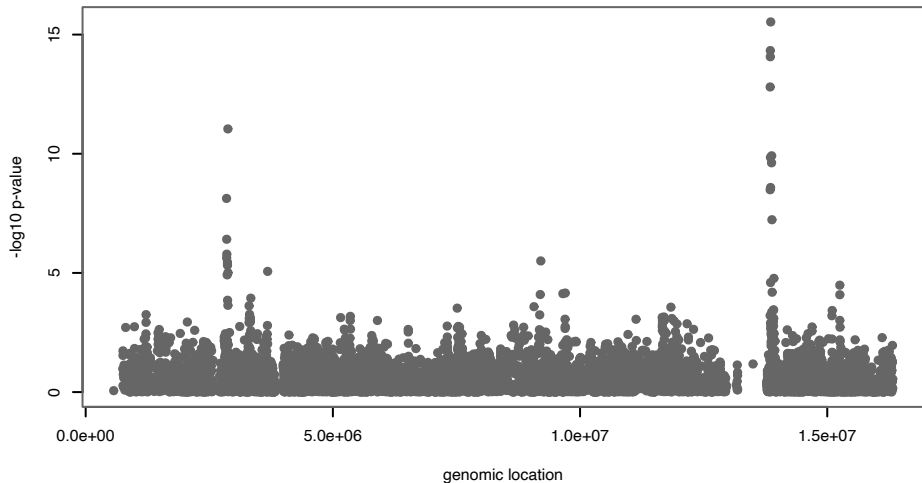
(Altshuler et al. 2008)

Human genetic variation recap

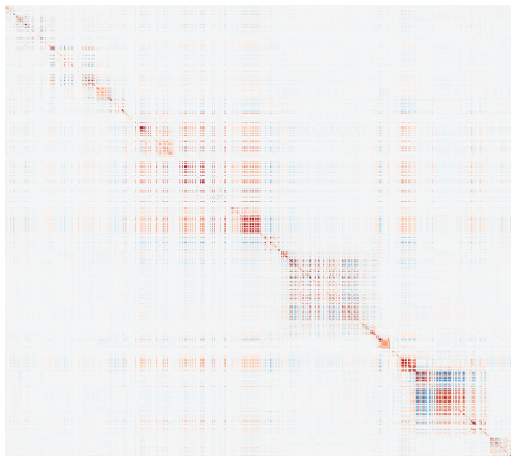


(Altshuler et al. 2008)

Manhattan plot: a summary of all the GWAS p-values



The covariance of X between the variants (columns)



For a genotype matrix X ($n \times p$),

Linkage disequilibrium matrix

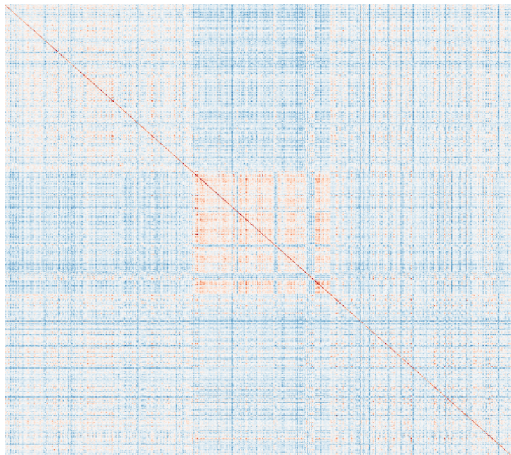
A $p \times p$ matrix:

$$\hat{R} = \frac{1}{n} X^T X$$

column $\mathbf{x}_j$ standardized (mean 0, SD 1).

$$R_{ij} = \frac{1}{n} \sum_{r=1}^n X_{ri} X_{rj}$$

The covariance of X between the individuals (rows)



For a genotype matrix X ($n \times p$)

Genetic relatedness matrix

An $n \times n$ matrix:

$$\hat{K} = \frac{1}{p} X X^T$$

each row $\mathbf{x}_i$ standardized (mean 0, SD 1).

$$K_{ij} = \frac{1}{p} \sum_{g=1}^p X_{ig} X_{jg}$$

What's next after GWAS?

- GWAS until 2010s: heavy focuses on **mapping**
 - GWAS map: genetic variants → a phenotype
 - Stringent genome-wide p-value cutoff
 - Study design, meta analysis
- NHGRI-EBI GWAS Catalog:
<https://www.ebi.ac.uk/gwas/>

Let's take a look at the GWAS Catalog:
<https://www.ebi.ac.uk/gwas/>

What's next after GWAS?

- GWAS since 2010s: more emphasis on **prediction**
 - Can we turn GWAS results to a prediction model?
 - Can we understand the mechanisms?
 - Machine learning, data integration, causal inference

Polygenic score models predict disease prevalence based on genome

- Poly + genic = many genes (units of hereditary components)

A (linear) polygenic score (PGS)

$$Y_i = \sum_j X_{ij} \beta_j$$

for an individual i .

Polygenic score models predict disease prevalence based on genome

- Poly + genic = many genes (units of hereditary components)

A (linear) polygenic score (PGS)

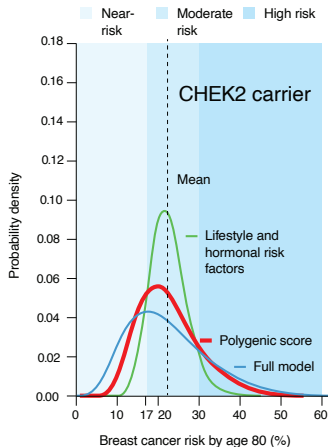
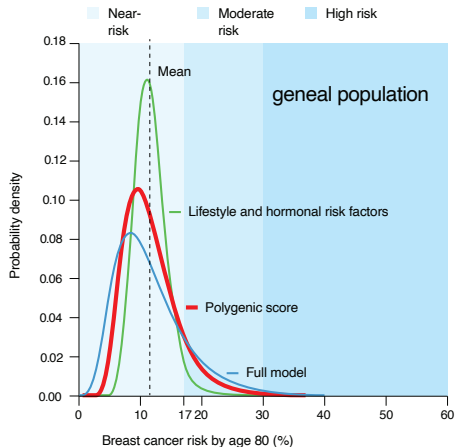
$$Y_i = \sum_j X_{ij} \beta_j$$

for an individual i .

Knowing β 's, we can predict:

$$Y_i^* \leftarrow \sum_j X_{ij}^* \hat{\beta}_j$$

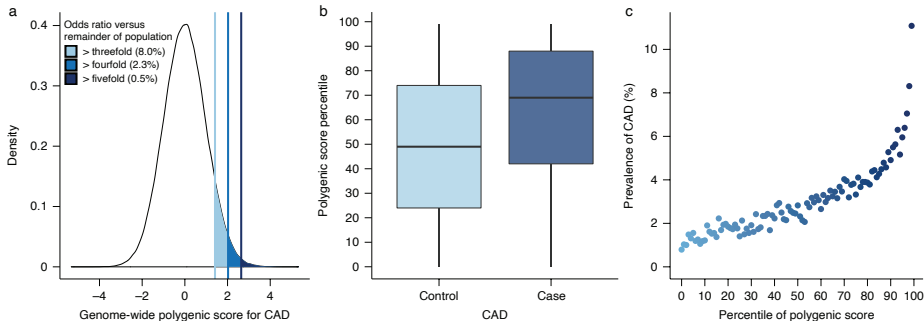
Example: PGS for breast cancer occurrence



- Strong heritability (proportion of disease risk variation explained by genetics): 35% - 80%
- Prediction is more accurate than lifestyle and other environmental factors
- More powerful if combined with rare risk factors

(Yang et al. 2023)

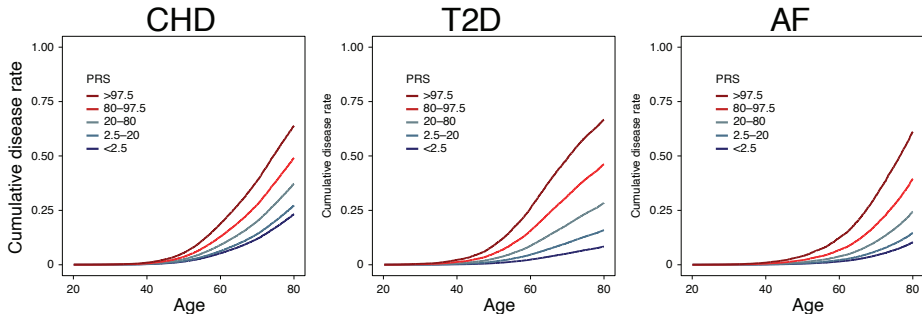
Example: Coronary artery disorder → prevention



- Top .5% of PGS values → five fold increase of CAD

(Khera et al. 2018)

Example: PGS stratifies individuals' disease onset

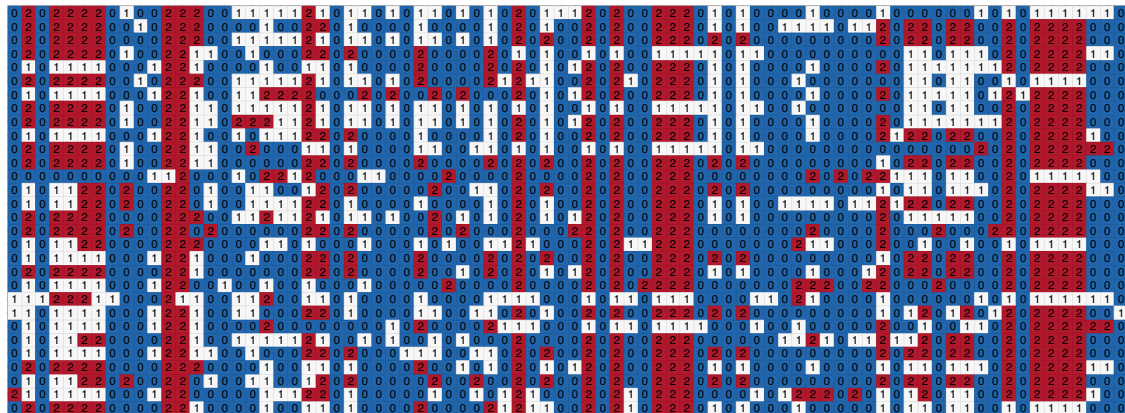


- $\text{PRS} \approx \text{PGS}$.
- We can partition cohorts based on PGS profiles.

(Mars et al. 2020)

GWAS for “Obsessive ggplot Disorder”

A genotype matrix X ($X_{ij} \in \{0, 1, 2\}$):



¹just for illustration purposes

In GWAS, we can have case vs. control phenotype

$Y_i = 1$ if case vs. $Y_i = 0$ if control

For our *OGD* GWAS:

```
table(y)
```

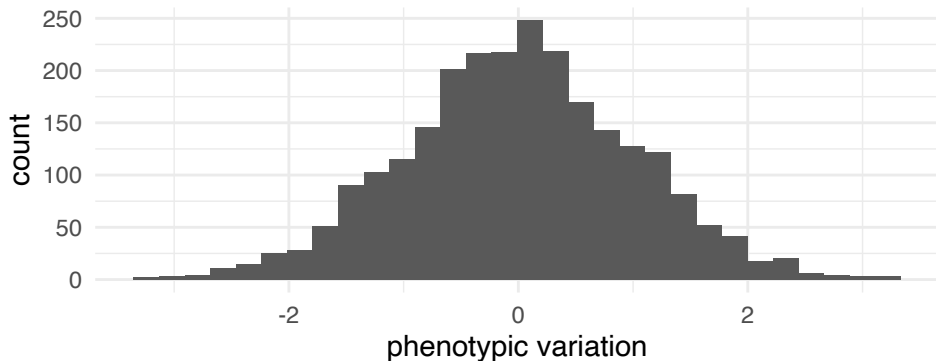
```
## y
```

```
## 0 1
```

```
## 731 769
```

- E.g., Cancer vs. non-cancer, schizophrenia vs. no mental disorder
- The fundamental question is, “Which individuals/subjects can be truly labelled control/wild type?”

Under the hood, there are quantitative “risk” scores



- E.g., height, BMI, parents' age of death, how many packs of cigarettes, etc.

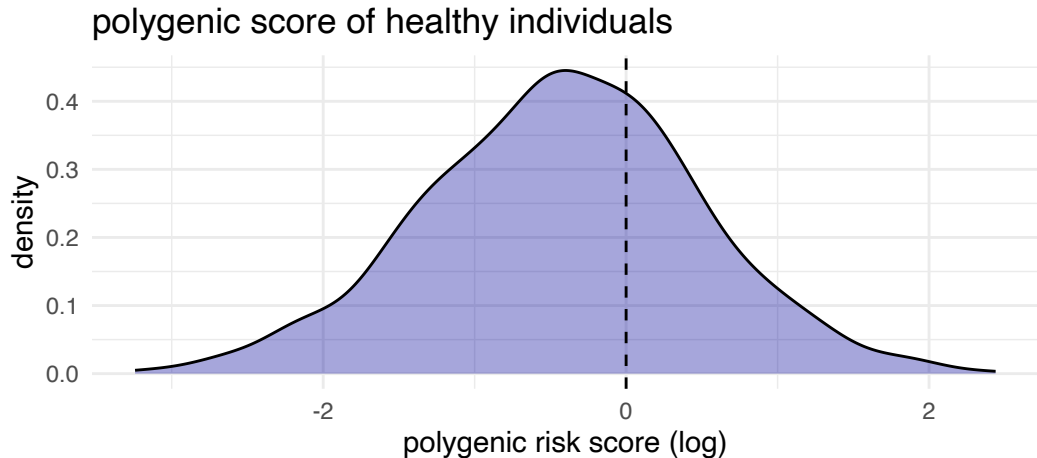
Side note: Almost all complex traits are polygenic

Polygenic effects of common SNPs → polymorphism in phenotypes

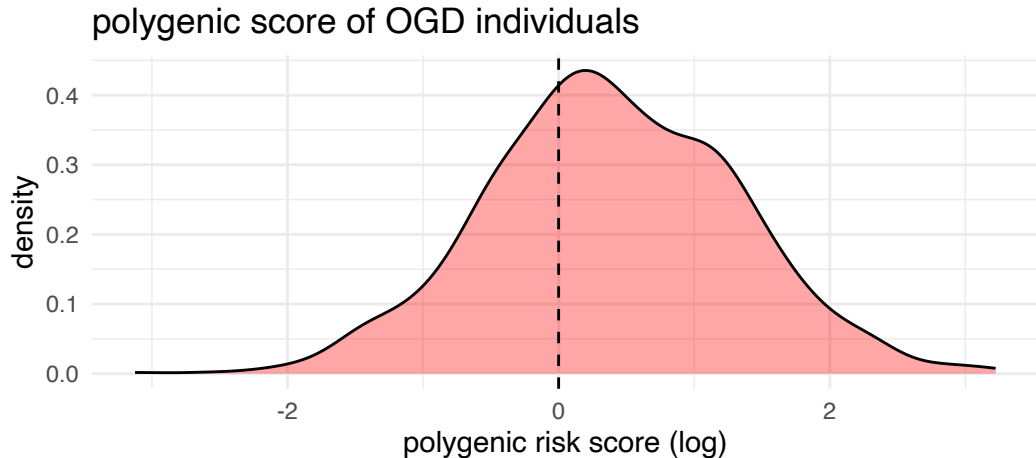
*A polygenic trait is a characteristic, such as height or skin color, that is influenced by two or more genes. **Because multiple genes are involved, polygenic traits do not follow the patterns of Mendelian inheritance.** Many polygenic traits are also influenced by the environment and are called multifactorial.*

<https://www.genome.gov/genetics-glossary/Polygenic-Trait>

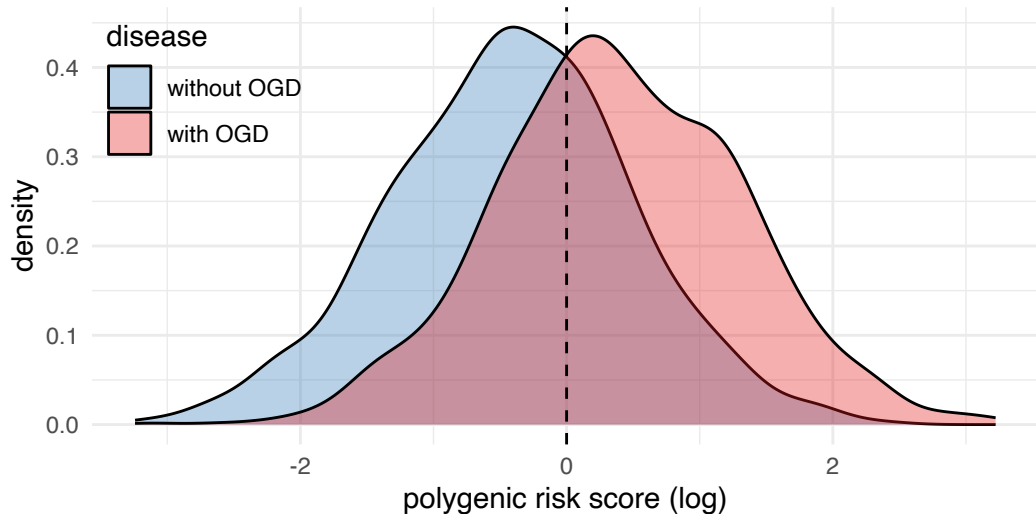
A polygenic score to predict disease prevalence



A polygenic score to predict disease prevalence



A polygenic score to predict disease prevalence



Polygenic score $\propto$ the odds of the case vs. control

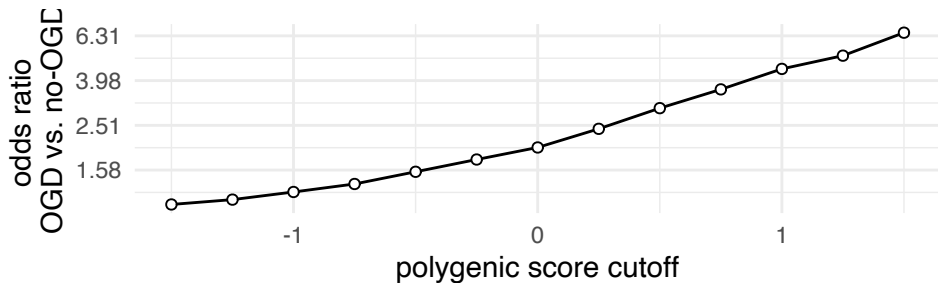
Log-odds ratio:

$$g(y) = \log \frac{p(\text{disease} | Y > y)}{p(\text{no disease} | Y > y)}$$

Polygenic score $\propto$ the odds of the case vs. control

Log-odds ratio:

$$g(y) = \log \frac{p(\text{disease} | Y > y)}{p(\text{no disease} | Y > y)}$$



A PGS model to explain disease prevalence

Log-odds ratio:

$$g(y) = \log \frac{p(\text{disease} | Y > y)}{p(\text{no disease} | Y > y)}$$

Goal: Estimate a function to predict this log-odds ratio.

$$g(y_i) \sim \beta_1 X_{i1} + \beta_2 X_{i2} \cdots + \epsilon$$

A PGS model to explain disease prevalence

Log-odds ratio:

$$g(y) = \log \frac{p(\text{disease} | Y > y)}{p(\text{no disease} | Y > y)}$$

Goal: Estimate a function to predict this log-odds ratio.

$$g(y_i) \sim \underbrace{\beta_1 X_{i1} + \beta_2 X_{i2} \cdots}_{\text{genetic effects}} + \epsilon$$

PGS estimation is a linear modelling with $p \gg n$

$$Y_i \sim X_{i1}\beta_1 + X_{i2}\beta_2 + \dots$$

PGS estimation is a linear modelling with $p \gg n$

$$\begin{pmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{pmatrix} \sim \begin{pmatrix} X_{11} \\ X_{21} \\ \vdots \\ X_{n1} \end{pmatrix} \beta_1 + \begin{pmatrix} X_{12} \\ X_{22} \\ \vdots \\ X_{n2} \end{pmatrix} \beta_2 + \dots$$

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$$\mathbf{y} \sim \mathbf{x}_1\beta_1 + \mathbf{x}_2\beta_2 + \cdots + \mathbf{x}_p\beta_p$$

$$p \gg n$$

PGS estimation is a linear modelling with $p \gg n$

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$$\mathbf{y} \sim \mathbf{x}_1\beta_1 + \mathbf{x}_2\beta_2 + \cdots + \mathbf{x}_p\beta_p$$

$$p \gg n$$

$n \approx 10^4$ but $p \approx 10^6$ for many large-scale GWAS

$p \gg n$: Why can't we just fit a model?

```
## lm.out <- lm(Y ~ X)
```

- $p \gg n$: need to estimate $\beta_1, \dots, \beta_p$

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```

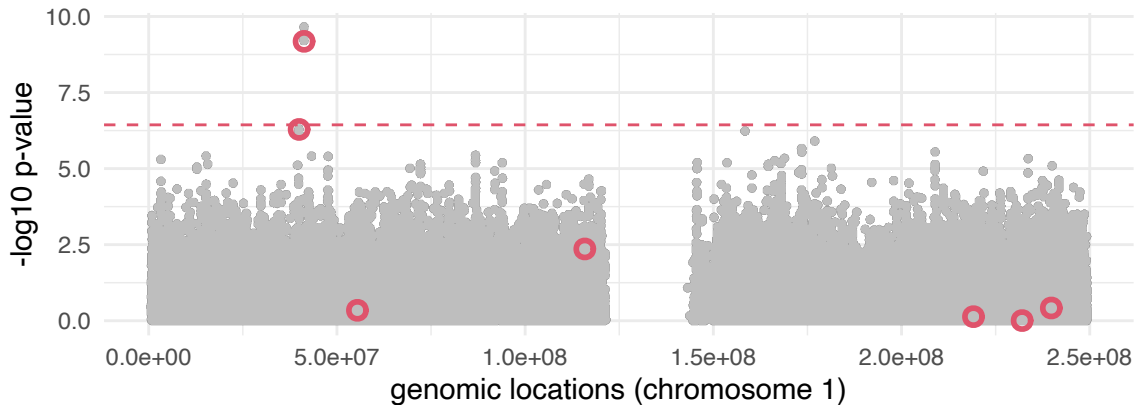
- $p \gg n$: need to estimate $\beta_1, \dots, \beta_p$
- How many samples? How many unknowns?
- Is it computationally feasible?

$p \gg n$: Variant-by-variant GWAS is not a bad idea

```
.gwas <- col_t_welch(X[Y == 0, ], X[Y == 1, ]) # univar.
```

$p \gg n$: Variant-by-variant GWAS is not a bad idea

```
.gwas <- col_t_welch(X[Y == 0, ], X[Y == 1, ]) # univar.
```



A linear model including only these “GWAS” variants

```
X.gwas <- X[, p.adjust(.gwas$pvalue) < .05]  
head(X.gwas, 3)
```

```
##           41270163 41270937 41272533 41274422  
## HG00159      0        0        0        0  
## HG01495      1        1        1        1  
## HG01519      2        2        2        2
```

```
lm.out <- lm(Y ~ X.gwas)
```

```
anova(lm.out)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: Y
```

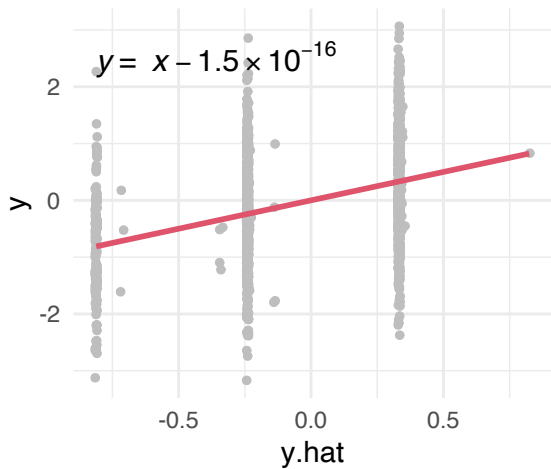
```
##           Df  Sum Sq Mean Sq F value    Pr(>F)  
## X.gwas      4   212.77   53.194  63.965 < 2.2e-16 ***
```

```
## Residuals 1495 1243.25    0.832
```

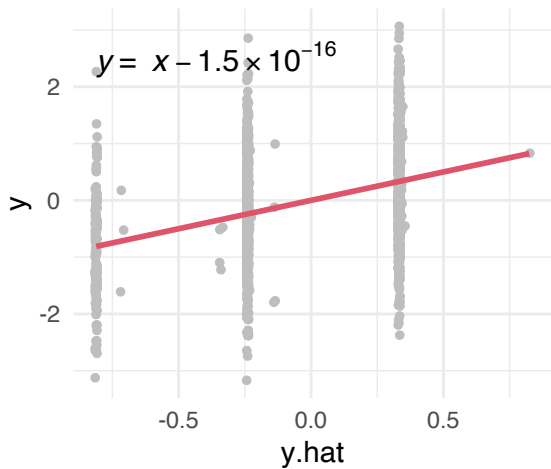
```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

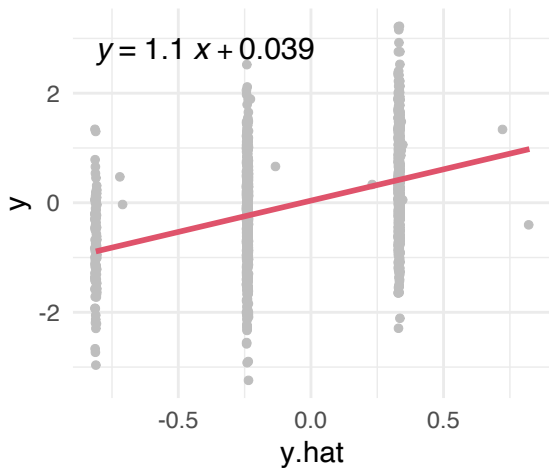
Train (N = 1500)



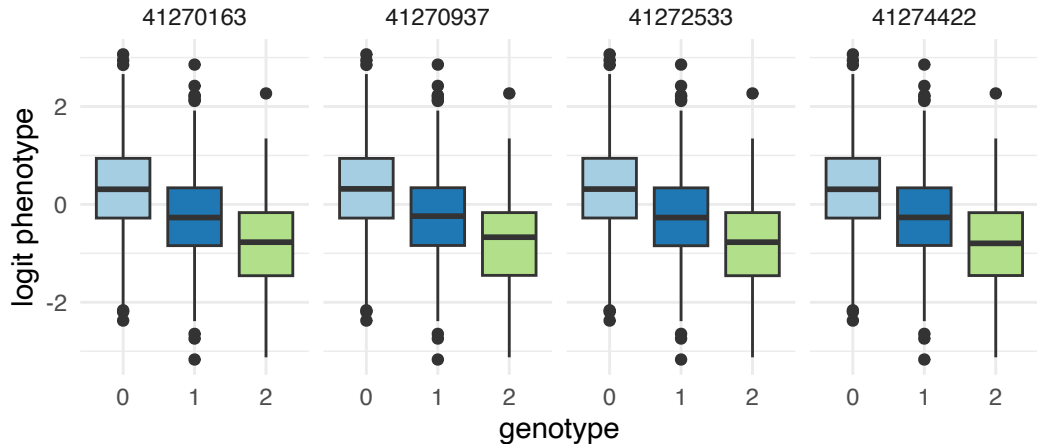
Train (N = 1500)



Test (N = 990)

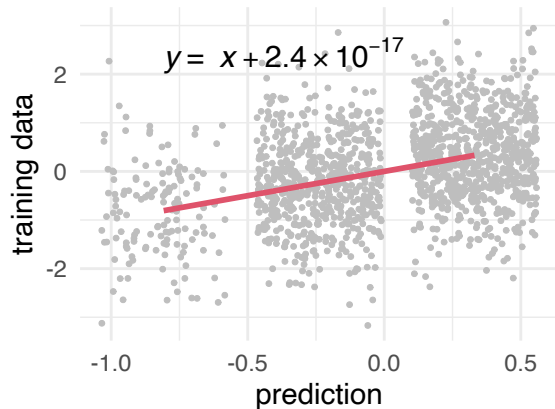


Do we need all of these GWAS variants?

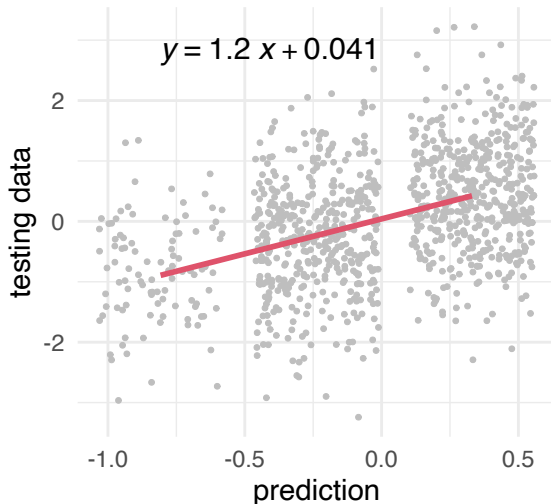


Just one of the GWAS variants works fine

Train (N = 1500)



Test (N = 990)



Simply add more variables?

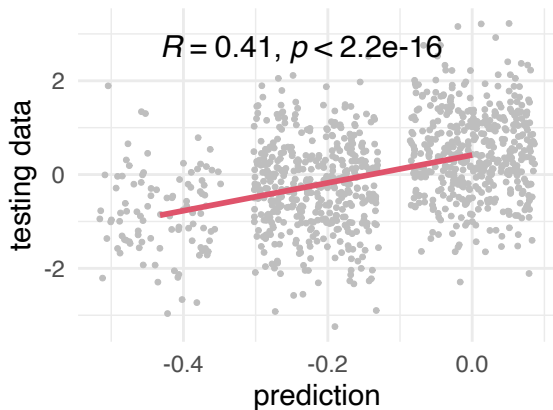
Let's do some experiments:

- 1 Pick top K variants (ordering by GWAS, $p_{(1)} < p_{(2)} < \dots$)
- 2 Take GWAS effect sizes β (the mean difference between case vs. control)
- 3 Predict by a linear combination of these effects:

$$\sum_{j=1}^K X_{i(j)} \beta_{(j)}$$

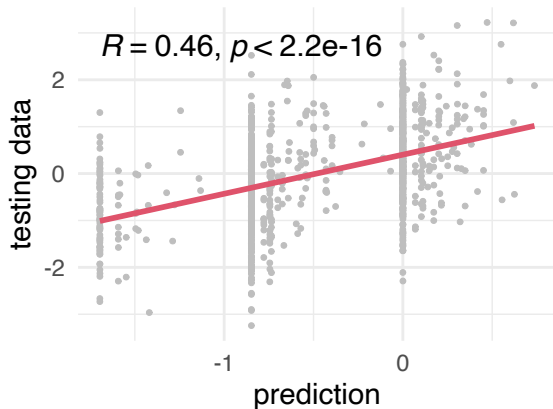
PGS is a variable selection problem

Top 1 variant



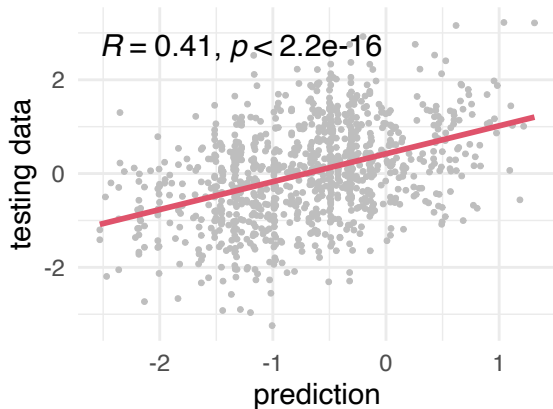
PGS is a variable selection problem

Top 10 variants



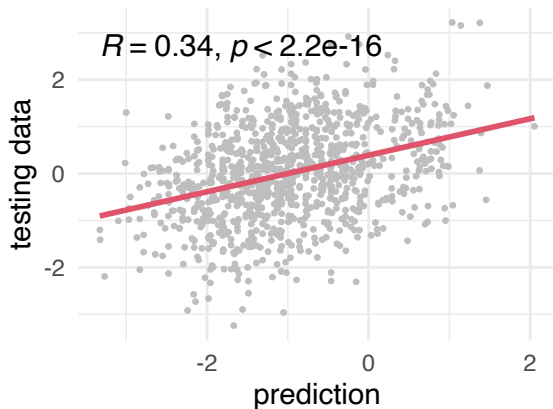
PGS is a variable selection problem

Top 20 variants



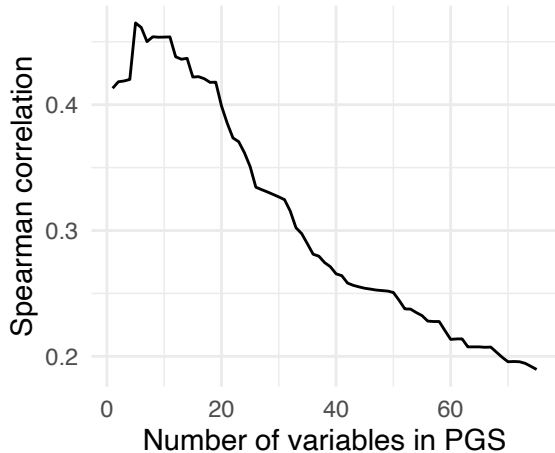
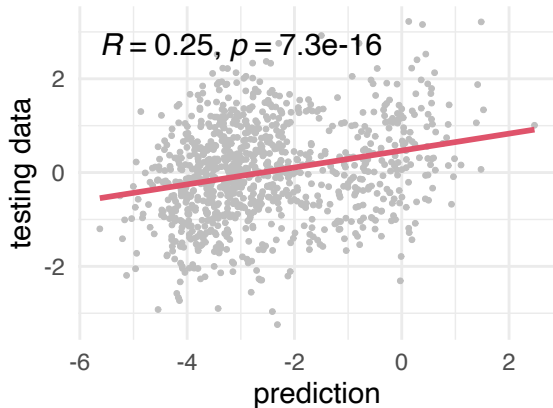
PGS is a variable selection problem

Top 30 variants



PGS is a variable selection problem

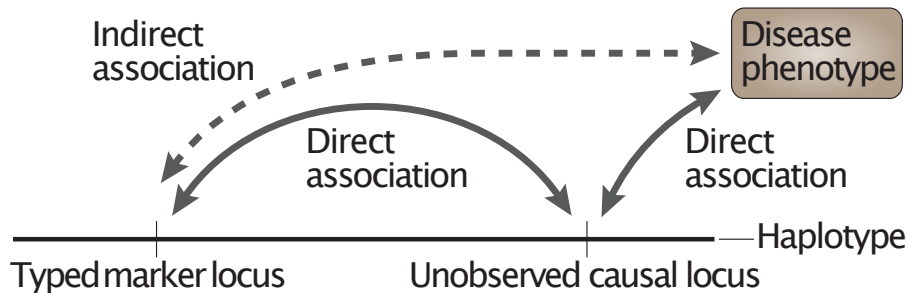
Top 50 variants



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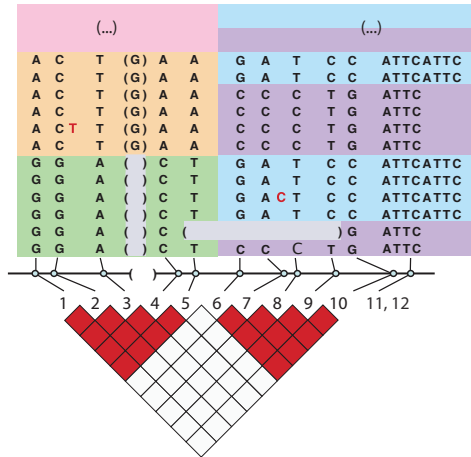
GWAS statistics only roughly tag causal variants



(Balding 2006)

Covariance between variants obfuscates GWAS interpretation

- Linkage disequilibrium (LD) block
- The result of recombination events throughout generations



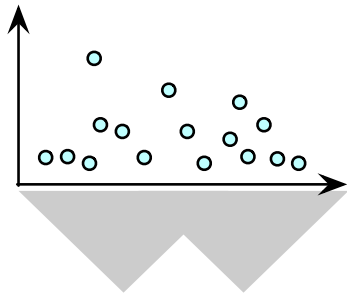
A typical setting of PGS training

PGS training

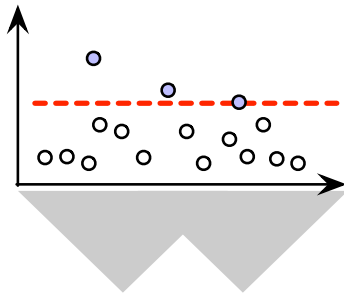
- Input:
 - ① GWAS summary statistics (p-values, univariate effect sizes)
 - ② $n \times p$ genotype matrix X , where $X_{ij} \in \{0, 1, 2\}$
 - ③ $n \times 1$ phenotype vector $\mathbf{y}$
- Output:
 - Polygenic effect sizes $\hat{\beta}_1, \dots, \hat{\beta}_p$
- Objective:
 - We want to predict unseen Y_i^* by $f(X^*; \hat{\beta})$ as accurately as possible
 - We want $\beta_j = 0$ as many as possible; only some $\beta_j \neq 0$.

Common strategies to deal with LD structures

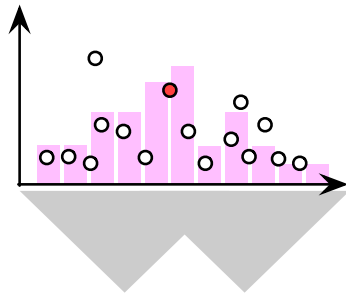
Strategy 1. Just use them all



Strategy 2. Pruning/clumping

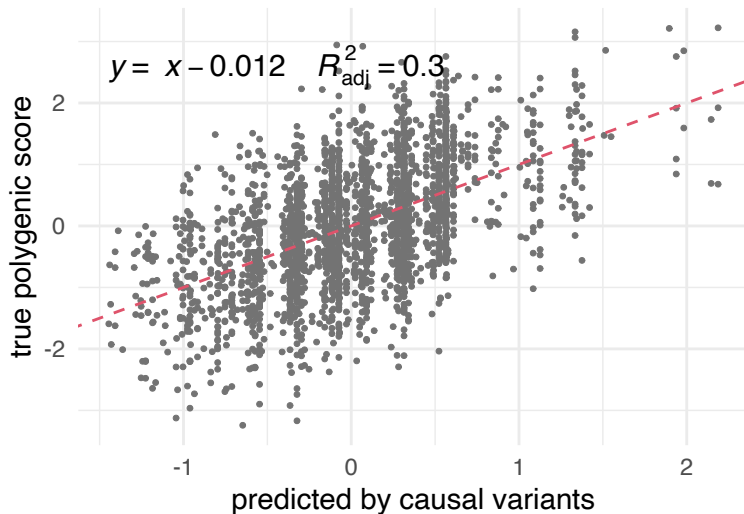


Strategy 3. **Fine-mapping**

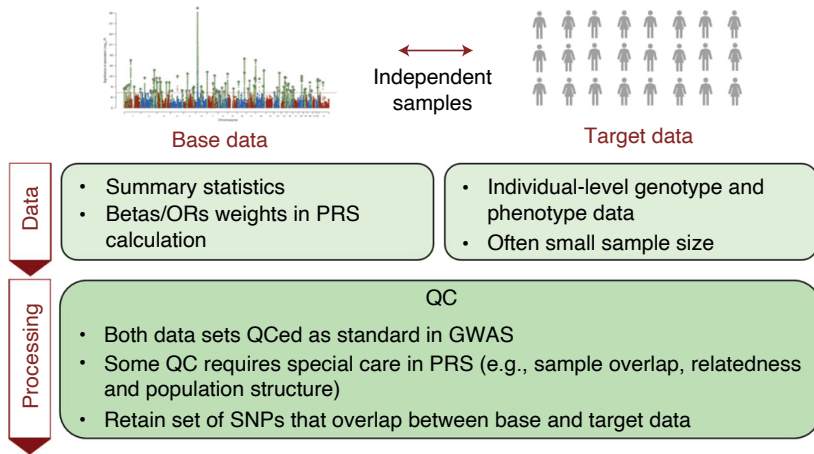


Causal variables explain a large fraction of var.

```
.lm <- lm(.sim$y.q ~ .sim$x[, .sim$causal, drop = FALSE] - 1)
```

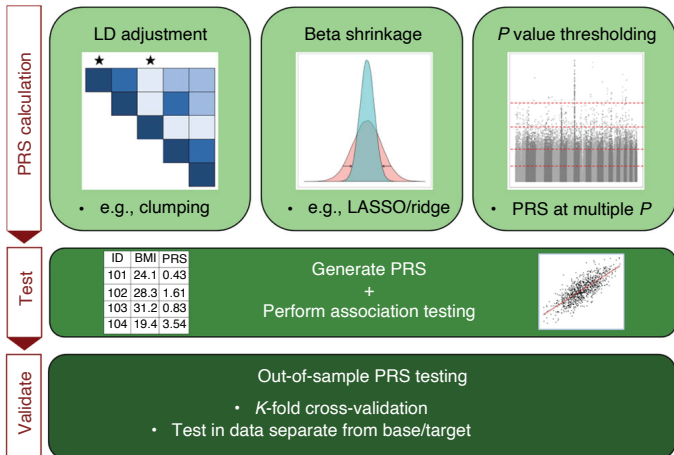


A general workflow of PGS estimation



(Choi et al. 2020)

A general workflow of PGS estimation



(Choi et al. 2020)

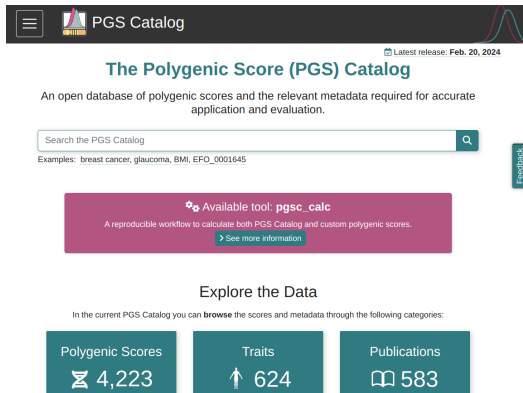
A proposed workflow of PGS estimation

- ① Take GWAS summary statistics and target population genotype X
- ② Run SuSiE² chromosome by chromosome to select variables
- ③ Predict PGS per chromosome and aggregate them

²SuSiE can run with summary statistics only

PGS Catalog: No need to train from scratch

<https://www.pgscatalog.org/>



The screenshot shows the PGS Catalog website. At the top, there is a dark header with a menu icon, the PGS Catalog logo, and a bell icon indicating the latest release date: Feb. 20, 2024. Below the header, the title "The Polygenic Score (PGS) Catalog" is displayed in a teal color. A subtitle reads: "An open database of polygenic scores and the relevant metadata required for accurate application and evaluation." A search bar with the placeholder text "Search the PGS Catalog" and a magnifying glass icon is present. Below the search bar, examples of search terms are listed: "breast cancer, glaucoma, BMI, EFO_0001645". A purple box highlights an available tool, "pgsc_calc", described as a reproducible workflow to calculate both PGS Catalog and custom polygenic scores, with a link to "See more information". Below this, the "Explore the Data" section states: "In the current PGS Catalog you can browse the scores and metadata through the following categories:". Three teal buttons are shown: "Polygenic Scores" with a DNA helix icon and the number 4,223; "Traits" with an upward arrow icon and the number 624; and "Publications" with a book icon and the number 583. A vertical "Feedback" button is located on the right side of the page.

PGS Catalog

Latest release: Feb. 20, 2024

The Polygenic Score (PGS) Catalog

An open database of polygenic scores and the relevant metadata required for accurate application and evaluation.

Search the PGS Catalog

Examples: breast cancer, glaucoma, BMI, EFO_0001645

Available tool: **pgsc_calc**

A reproducible workflow to calculate both PGS Catalog and custom polygenic scores.

[See more information](#)

Explore the Data

In the current PGS Catalog you can **browse** the scores and metadata through the following categories:

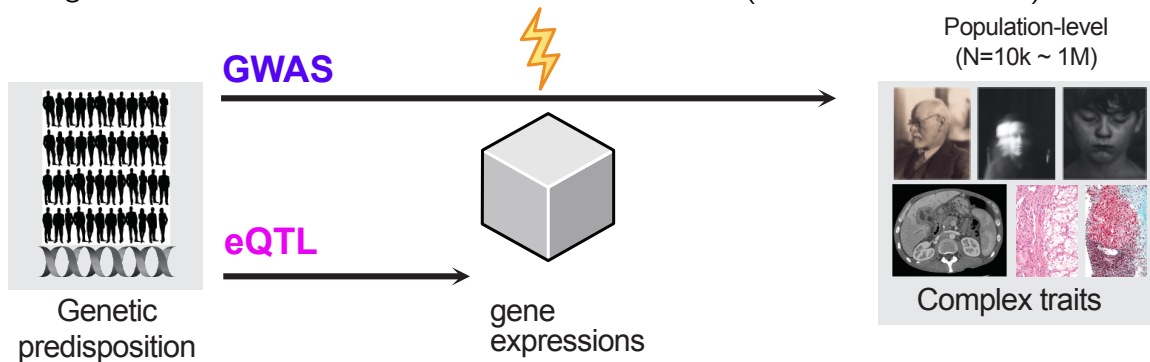
Polygenic Scores	Traits	Publications
4,223	624	583

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- 2 Fundamental challenges in PGS (regression)
- 3 Expression Quantitative Trait Loci ($G \rightarrow \text{Expr.}$)**
- 4 Co-localization in genomics

Transcriptome-wide association study (TWAS)

Using eQTL, we can understand GWAS mechanisms... (Gamazon et al. 2015)



$$\mathbf{m}_g \sim X\boldsymbol{\alpha}_g + \boldsymbol{\epsilon}_g \quad \implies \quad \mathbf{y}_{\text{GWAS}} = \sum_{g \in \text{causal genes}} \mathbf{m}_g \beta_g + \boldsymbol{\epsilon}_y$$

eQTL mapping as a linear regression problem

A variant-by-variant association test for a variant j :

$$H_0 : \quad \mathbb{E}[Y_i|X_{ij}] = \mu$$

vs.

$$H_1 : \quad \mathbb{E}[Y_i|X_{ij}] = \mu + X_{ij}\beta_j$$

- For gene expression Y , test association with each variant X_j
- Model: $Y_i = X_{ij}\beta_j + \epsilon$
- Repeat for all gene-variant pairs in cis window (e.g., $\pm 1\text{Mb}$)

The high-dimensional challenge: $p \gg n$

Predicting expression from genotypes is a high-dimensional problem:

$$Y_i = \sum_{j=1}^p X_{ij} \theta_j + \epsilon$$

- $p \gg n$: need to estimate $\theta_1, \dots, \theta_p$

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- $p \gg n$: need to estimate $\theta_1, \dots, \theta_p$
- How many samples? How many unknowns?
- Is it computationally feasible?
- **Standard linear regression fails!**

Penalized regression: Motivation

- Many predictors (variants), few samples
- Need to select relevant variants
- Avoid overfitting
- Two main approaches:
 - Ridge (L2): shrinkage without variable selection
 - LASSO (L1): shrinkage + sparse selection
 - Elastic Net: combination of both

Application: PrediXcan framework

Goal: Predict gene expression from genotypes to understand GWAS

Two-stage approach:

- ① **Training stage:** Build expression prediction model

$$\hat{E}_g = \sum_j X_j \beta_j^{(g)}$$

using LASSO/Elastic Net on eQTL reference panel

- ② **Association stage:** Test predicted expression with trait

$$Y \sim \hat{E}_g + \text{covariates}$$

Why penalized regression? Need to select causal eQTL variants from many correlated

Lasso, a linear regression with Laplace prior (L1)

Prior distribution

$$p(\boldsymbol{\theta}) = \text{Laplace}(\boldsymbol{\theta}|\lambda) \propto \exp(-\lambda\|\boldsymbol{\theta}\|_1)$$

where

$$\|\boldsymbol{\theta}\|_1 = \sum_{j=1}^p |\theta_j|, \text{ L1-norm.}$$

Maximize

$$\ln p(\mathbf{y}|X, \boldsymbol{\theta}) + \ln p(\boldsymbol{\theta}|\lambda) = -\frac{1}{2\sigma^2} \sum_{i=1}^n (y_i - \mathbf{x}_i\boldsymbol{\theta})^2 - \lambda\|\boldsymbol{\theta}\|_1$$

Minimize L_1 -regularized error

$$\sum_{i=1}^n (y_i - \mathbf{x}_i\boldsymbol{\theta})^2 + \lambda\|\boldsymbol{\theta}\|_1$$

glmnet solves this regularized optimization problem

Goal (by variable-by-variable updates):

$$\min_{\beta} \quad \overbrace{(\mathbf{y} - X\beta)^\top (\mathbf{y} - X\beta)}^{\text{RSS}} + \underbrace{\lambda\alpha\|\beta\|_1}_{\text{variable selection}} + \underbrace{\lambda(1-\alpha)\|\beta\|_2}_{\text{shrinkage}}$$

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For each β_j ,

$$\hat{\beta}_j^{\text{glmnet}} \leftarrow \frac{S\left(\sum_{i=1}^n X_{ij}(y_i - \hat{y}_i^{(-j)}), \lambda\alpha\right)}{\sum_{i=1}^n X_{ij}^2 + \lambda(1-\alpha)}$$

Friedman *et al.*, Regularization Paths for Generalized Linear Models via Coordinate Descent (2010)

glmnet solves this regularized optimization problem

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$$\min_{\beta} \quad \overbrace{(\mathbf{y} - X\beta)^\top (\mathbf{y} - X\beta)}^{\text{RSS}} + \underbrace{\lambda\alpha\|\beta\|_1}_{\text{variable selection}} + \underbrace{\lambda(1-\alpha)\|\beta\|_2}_{\text{shrinkage}}$$

For each β_j ,

$$\hat{\beta}_j \leftarrow \frac{\overset{\text{threshold}}{S} \left(\overset{\text{residual w/o the variable } \beta_j}{\sum_{i=1}^n X_{ij} \quad (y_i - y_i^{(-j)})}, \lambda\alpha \right)}{\sum_{i=1}^n X_{ij}^2 + \underbrace{\lambda(1-\alpha)}_{\text{shrinkage}}}$$

where $S(z, \tau)$ will set it to zero if $|z| < \tau$.

Sum of Single Effect (SuSiE) regression

```
library(susieR)
Y.train <- .sim$y.q[.train]
susie.out <- susie(X, Y.train)
```

$$\mathbf{y} = \sum_{l=1}^L \underbrace{\sum_j \mathbf{x}_j \overset{\text{probabilistic selection}}{\alpha_j^{(l)}} \overset{\text{single variant effect}}{\beta_j^{(l)}}}_{\text{layer-by-layer}} + \epsilon$$

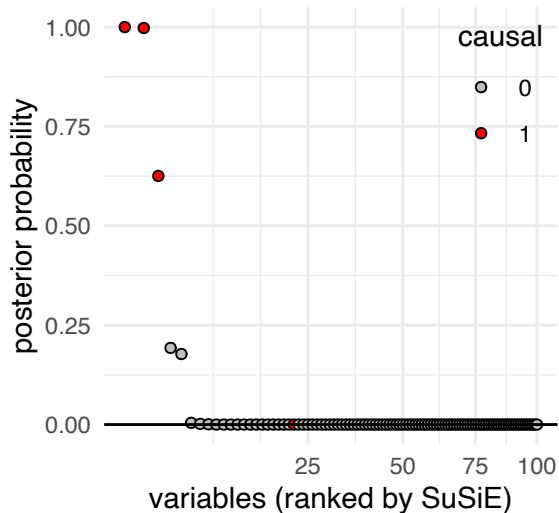
where $\sum_{j=1}^p \alpha_j^{(l)} = 1$ for each layer l .

Wang .. Stephens, *Journal of the Royal Statistical Society* (2020)

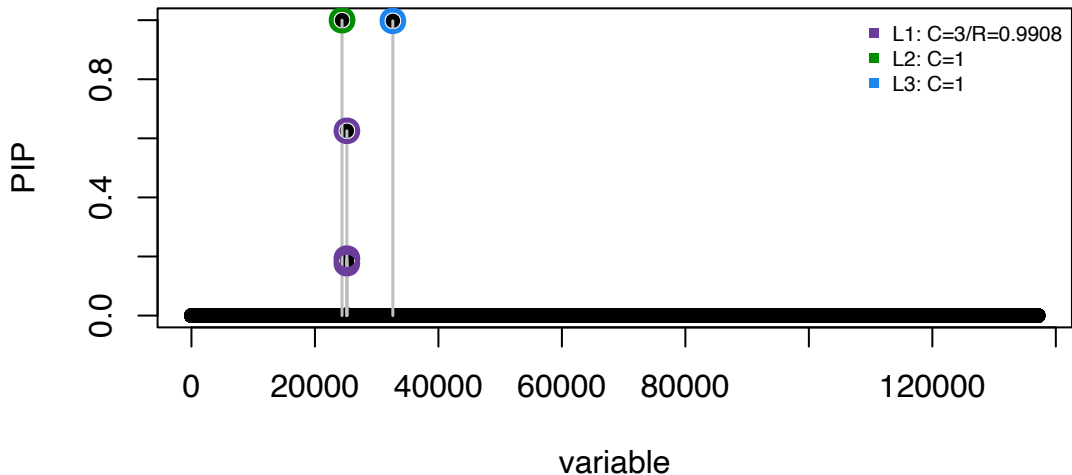
Ideas behind SuSiE

- ① Estimating univariate effects is easy (GWAS)
- ② Many variants can show similar effects (LD)
- ③ Let's weight variants probabilistically
- ④ Regress out the probabilistically reweighted effects
- ⑤ Repeat 1-4.

SuSiE identifies top causal variants

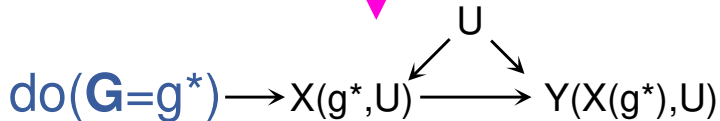
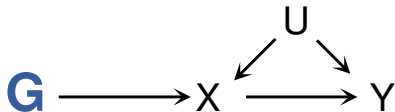


SuSiE provides credible sets



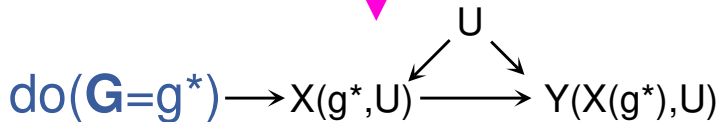
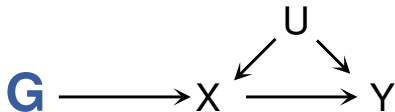
Credible sets: Groups of variants that contain the causal variant with high probability (e.g., 95%)

Mendelian Randomization



We will use a genetic variable G to mimic an RCT

Mendelian Randomization



If a genetic variable G is a valid instrumental variable...

MR in modern epidemiology studies

- G: genotype
- X: APOE protein
- Y: cancer

$$G \rightarrow X \rightarrow Y$$

APOLIPOPROTEIN E ISOFORMS, SERUM CHOLESTEROL, AND CANCER

SIR,—It is unclear whether the relation between low serum cholesterol levels and cancer¹ is causal. In many studies occult tumour may have depressed cholesterol levels though in others the relation was found when serum cholesterol had been measured many years before the cancer was diagnosed. The relation is probably not explained by diet, because in the Seven Countries Study cohorts with widely different diets and corresponding differences in mean cholesterol levels experienced similar mean cancer rates.^{2,3} On the other hand, within each region cancer incidence was higher in men with a serum cholesterol in the lowest part of the cholesterol distribution for that country.³ Thus, naturally low cholesterol levels are sometimes associated with increased cancer risk.^{1,3}

Differences in the aminoacid sequence of apolipoprotein E (apo

Katan, *Lancet*, (1986)

MR in modern epidemiology studies

- G: genotype
- X: APOE protein
- Y: cancer

$$G \rightarrow X \rightarrow Y$$



30TH THOMAS FRANCIS JR MEMORIAL LECTURE

'Mendelian randomization': can genetic epidemiology contribute to understanding environmental determinants of disease?*

George
Davey Smith

George Davey Smith and Shah Ebrahim

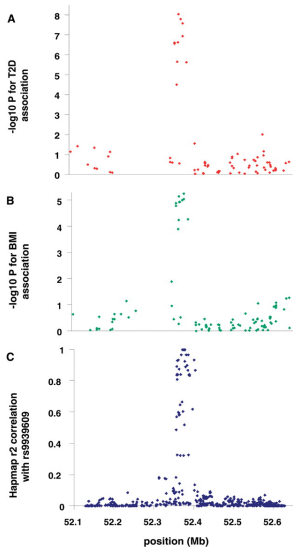
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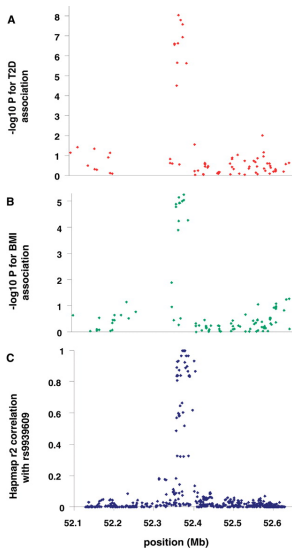
MR in action: *FTO* → fat mass → obesity, diabetes



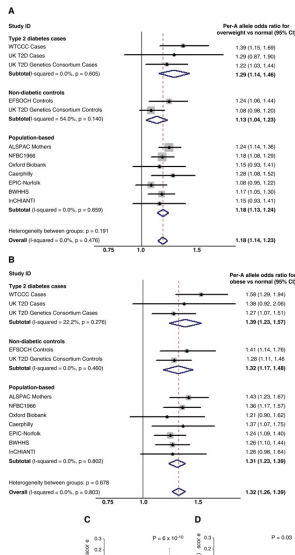
Using genotype as an instrumental variable, we test causality between other exposure variables and downstream phenotypes.

● Genotype in *FTO* locus → T2D

MR in action: *FTO* → fat mass → obesity, diabetes



Yongjin Park

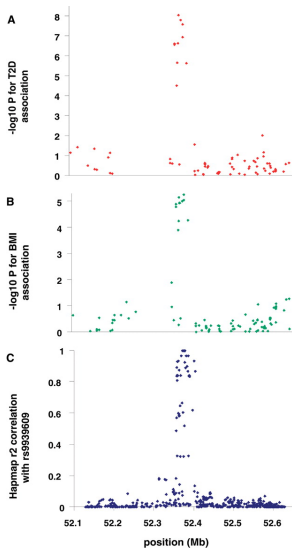


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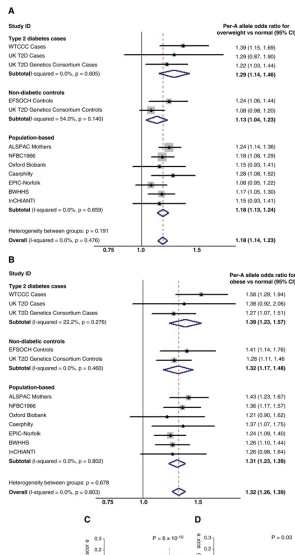
- Genotype in *FTO* locus → T2D
- *FTO* locus → fat mass

25 February 2026

MR in action: *FTO* → fat mass → obesity, diabetes



Yongjin Park



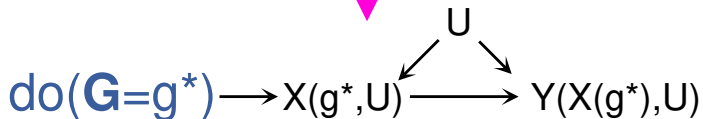
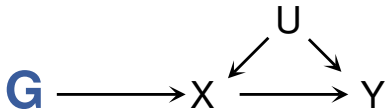
Regression for Regulatory Genomics

Using genotype as an instrumental variable, we test causality between other exposure variables and downstream phenotypes.

- Genotype in *FTO* locus → T2D
- *FTO* locus → fat mass
- Using *FTO* as "instrumental variable", we can ask other MR questions

25 February 2026

Why use Mendelian Randomization?



- ① We do not have enough knowledge about U
- ② We do not have a way to intervene on X , i.e., $\text{do}(X = x)$

“Genotypes are beautifully randomized” (Fisher 1952)

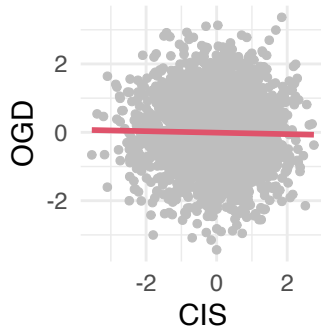
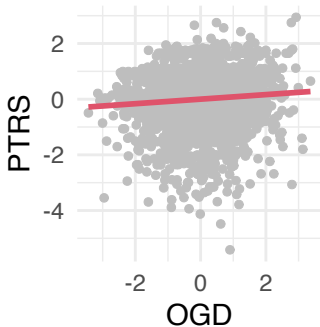
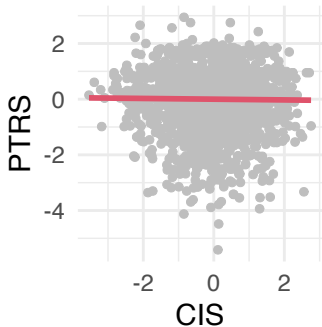
How can we learn causality between genetic disorders?

Three disorders (just for demonstration)

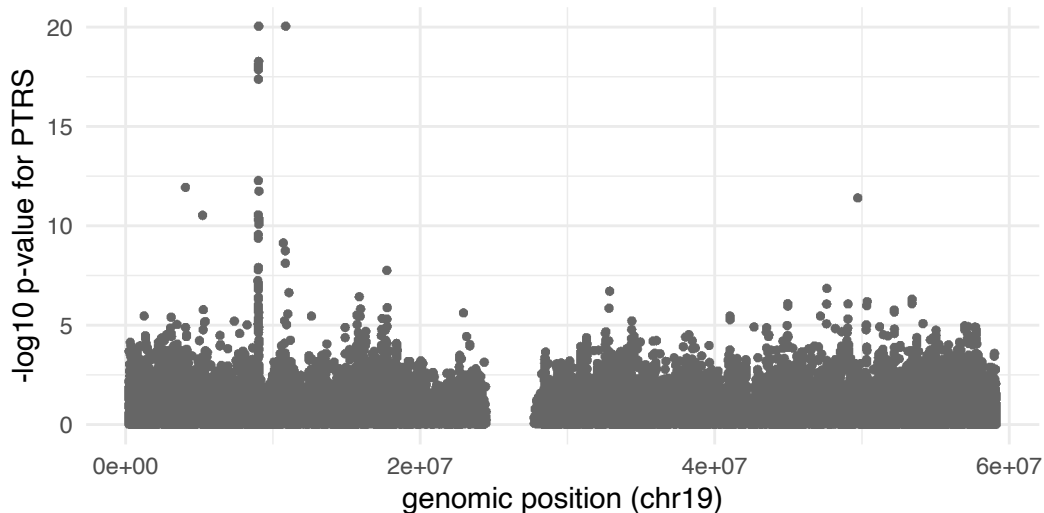
- OGD: Obsessive `ggplot` disorder
- PTRS: Post-traumatic R session stress syndrome
- CIS: Chronic indentation syndrome (as a result of Python coding)

How can we learn causality between genetic traits?

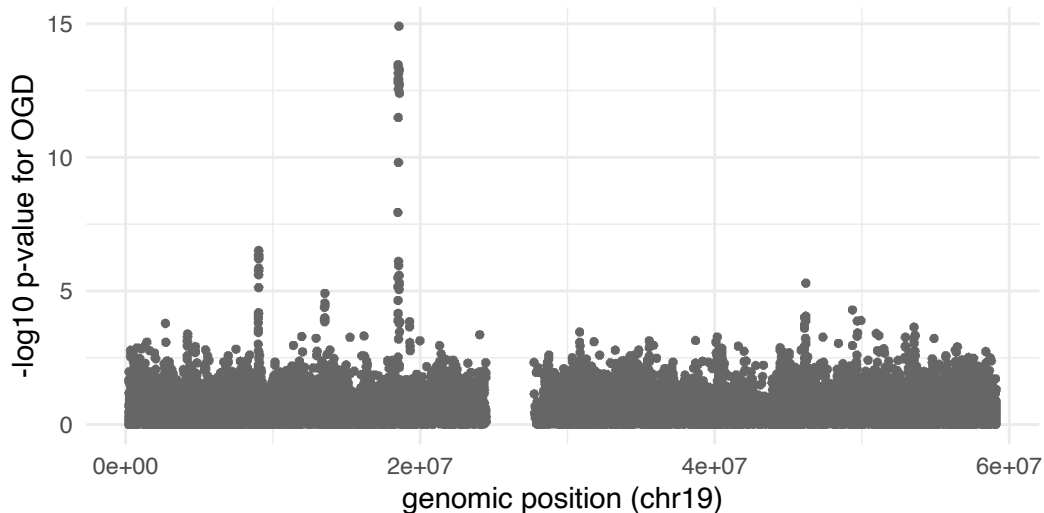
In previous longitudinal studies, we found the following relationships.



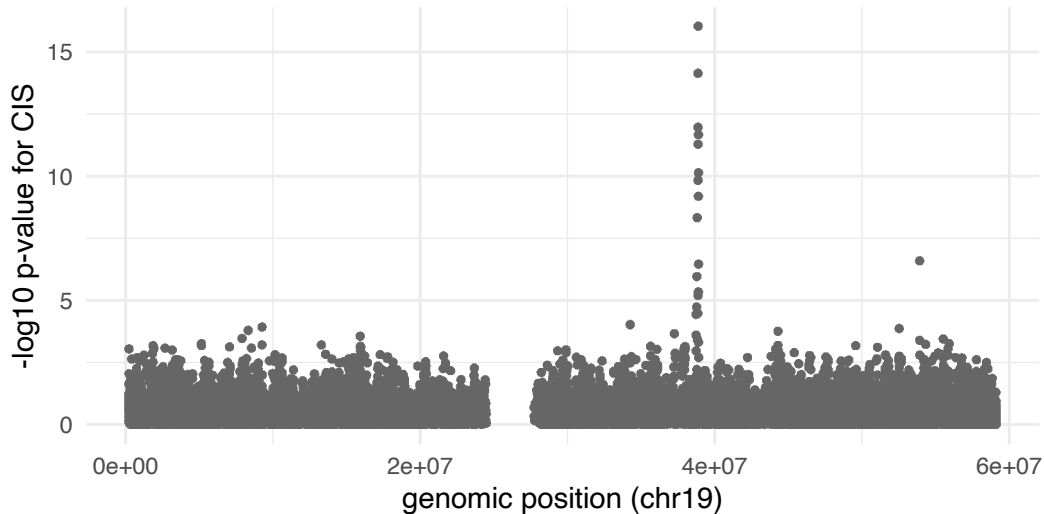
Can we recover such relationships from GWAS data?



Can we recover such relationships from GWAS data?



Can we recover such relationships from GWAS data?



MR measures mediation effects of X to Y

Given that G is a valid instrumental variable...

Example:

- X : gene expression
- Y : disease phenotype

Suppose we have estimated

$$\begin{aligned} G &\overset{\alpha}{\rightarrow} X \\ X &= G\hat{\alpha} + \epsilon_X \end{aligned}$$

MR measures mediation effects of X to Y

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- X : gene expression
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Suppose we have estimated

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and

$$G \xrightarrow{\gamma} Y$$

MR measures mediation effects of X to Y

Given that G is a valid instrumental variable...

Example:

- X : gene expression
- Y : disease phenotype

Goal: What is the causal effect β in $X \xrightarrow{\beta} Y$?

$$Y = X\beta + \epsilon'$$

Suppose we have estimated

$$G \xrightarrow{\alpha} X$$

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and

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MR measures mediation effects of X to Y

Given that G is a valid instrumental variable...

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- X : gene expression
- Y : disease phenotype

Suppose we have estimated

Goal: What is the causal effect β in $X \xrightarrow{\beta} Y$?

$$\begin{aligned} Y &= X\beta + \epsilon' \\ G\hat{\gamma} + \epsilon_Y &= (G\hat{\alpha} + \epsilon_X)\beta + \epsilon' \end{aligned}$$

$$\begin{aligned} G &\xrightarrow{\alpha} X \\ X &= G\hat{\alpha} + \epsilon_X \end{aligned}$$

and

$$G \xrightarrow{\gamma} Y$$

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$$G \overset{\gamma}{\rightarrow} Y$$

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$$\begin{aligned} Y &= X\beta + \epsilon' \\ G\hat{\gamma} + \epsilon_Y &= (G\hat{\alpha} + \epsilon_X)\beta + \epsilon' \\ G\hat{\gamma} &= G\hat{\alpha}\beta + \dots \end{aligned}$$

MR measures mediation effects of X to Y

Given that G is a valid instrumental variable...

Example:

- X : gene expression
- Y : disease phenotype

Suppose we have estimated

$$G \xrightarrow{\alpha} X$$

$$X = G\hat{\alpha} + \epsilon_X$$

and

$$G \xrightarrow{\gamma} Y$$

Goal: What is the causal effect β in $X \xrightarrow{\beta} Y$?

$$Y = X\beta + \epsilon'$$

$$G\hat{\gamma} + \epsilon_Y = (G\hat{\alpha} + \epsilon_X)\beta + \epsilon'$$

$$G\hat{\gamma} = G\hat{\alpha}\beta + \dots$$

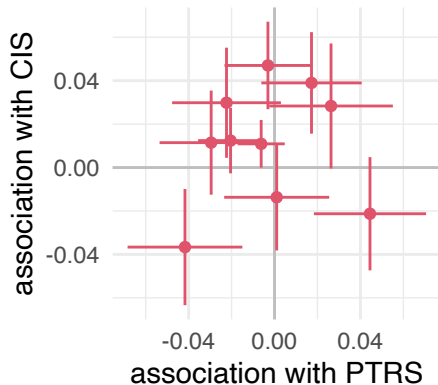
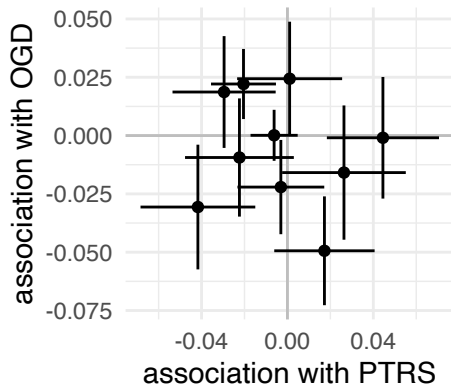
The answer is as simple as

$$\mathbb{E}[\beta] = \frac{\gamma}{\alpha}$$

We will revisit this problem in the GWAS

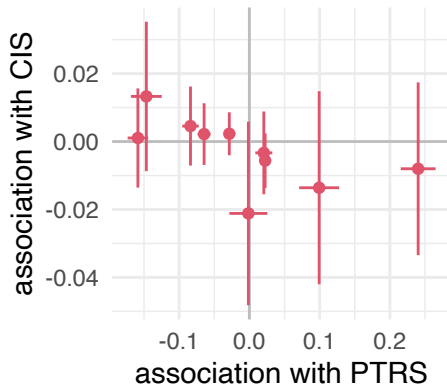
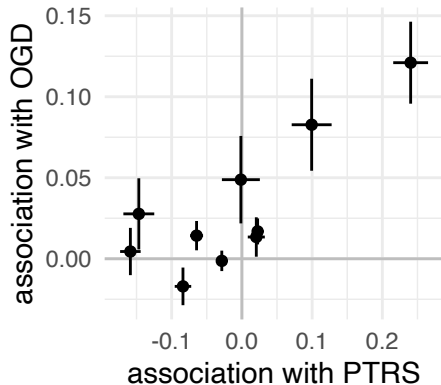
Will any genetic variable work for MR analysis?

Goal: Is PTRS $\rightarrow$ OGD or PTRS $\rightarrow$ CIS?

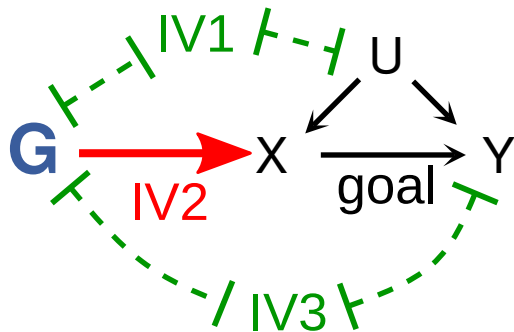


Select causal variants of exposure (PTRS)?

Goal: Is $\text{PTRS} \rightarrow \text{OGD}$ or $\text{PTRS} \rightarrow \text{CIS}$?



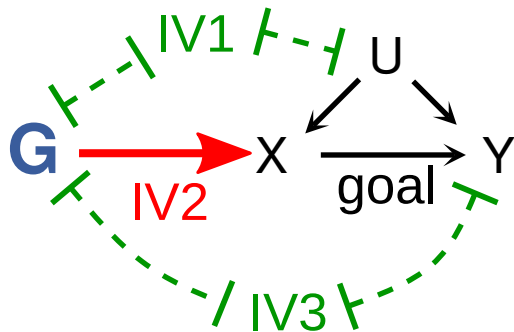
Definition of instrumental variable in MR study



- IV1: The genetic variant G is independent of the potential confounder variable U

(Bowden et al. 2015)

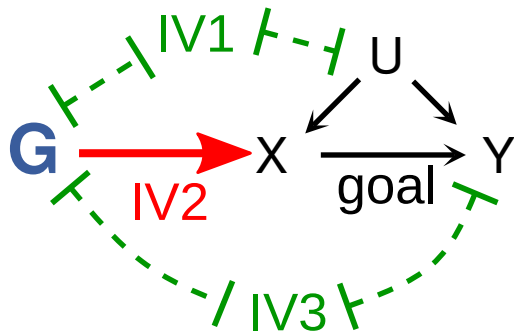
Definition of instrumental variable in MR study



- IV1: The genetic variant G is independent of the potential confounder variable U
- IV2: The genetic variant is associated with the exposure X

(Bowden et al. 2015)

Definition of instrumental variable in MR study



- IV1: The genetic variant G is independent of the potential confounder variable U
- IV2: The genetic variant is associated with the exposure X
- IV3: The genetic variant is independent of the outcome Y conditioning on X

(Bowden et al. 2015)

Today's lecture

- 1 Polygenic risk prediction ($G \rightarrow \text{phenotype}$)
- 2 Fundamental challenges in PGS (regression)
- 3 Expression Quantitative Trait Loci ($G \rightarrow \text{Expr.}$)
- 4 Co-localization in genomics**

What is co-localization?

Co-localization: Two traits share the same causal variant(s) in a genomic region

- If two GWAS signals co-localize, they likely share a causal variant
- Important for: MR validation, functional follow-up, understanding pleiotropy

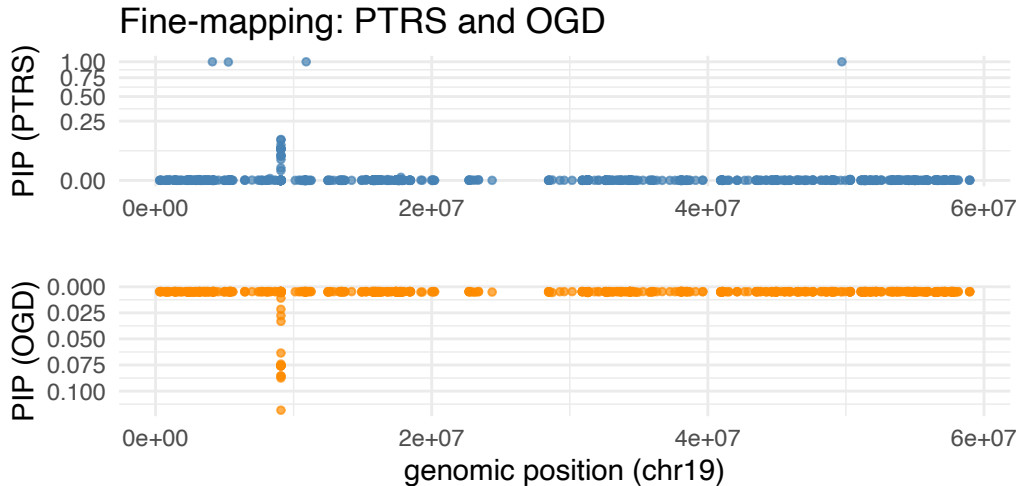
Our simulation:

- PTRS and OGD share 2 causal variants (co-localized)

Fine-mapping with SuSiE

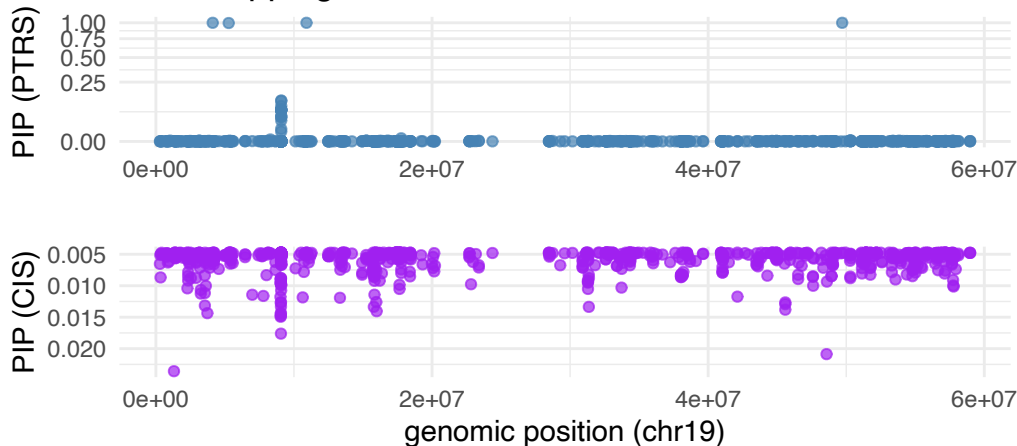
- SuSiE (Sum of Single Effects) identifies causal variants
- Outputs PIP (Posterior Inclusion Probability) for each variant
- High PIP = likely to be causal

Co-localization: PTRS and OGD (mediated)



No co-localization: PTRS and CIS (independent)

Fine-mapping: PTRS and CIS have different causal variants



Statistical test: coloc with SuSiE

Using coloc.susie (Wallace 2021) to test for co-localization:

```
## PTRS vs OGD (should co-localize)
```

```
coloc.pters.ogd$summary
```

```
##      nsnps      hit1      hit2      PP.H0.abf      PP.H1.abf      PP.H2.abf PP.H3.abf
##      <int>    <char>    <char>          <num>          <num>          <num>    <num>
## 1:    841 10907286 9078190 1.688239e-21 0.0211344301 7.817507e-20 0.9786441
## 2:    841 49705291 9078190 2.045804e-06 0.0211225288 9.473239e-05 0.9780921
## 3:    841 5273571 9078190 1.436645e-04 0.0209848186 6.652486e-03 0.9717157
## 4:    841 4115684 9078190 7.781789e-08 0.0211264781 3.603412e-06 0.9782751
## 5:    841 9079464 9078190 9.825459e-22 0.0006701868 4.549747e-20 0.0290930
##      PP.H4.abf  idx1  idx2
##      <num> <int> <int>
## 1: 0.0002214617     1     1
## 2: 0.0006886163     3     1
## 3: 0.0005033485     4     1
## 4: 0.0005947005     5     1
## 5: 0.9702368166     2     1
```

```
## PTRS vs CIS (should NOT co-localize)
```

```
coloc.pters.cis$summary
```

Interpreting coloc results

Posterior probabilities (PP) (Wallace 2021):

- **PP.H0**: No association in either trait
- **PP.H1**: Association with trait 1 only
- **PP.H2**: Association with trait 2 only
- **PP.H3**: Both associated, different causal variants
- **PP.H4**: Both associated, **shared causal variant** (co-localization)

Rule of thumb: $PP.H4 > 0.8$ suggests strong evidence for co-localization

coloc.susie allows for multiple causal variants per trait

Summary: Visual + Statistical evidence

Co-localized (PTRS-OGD):

- High PIPs at same positions (mirror image)
- High PP.H4 from coloc.susie
- Suggests shared causal variant

Not co-localized (PTRS-CIS):

- High PIPs at different positions
- Low PP.H4, higher PP.H3
- Independent causal variants

Working with GWAS summary statistics

Often we only have access to GWAS summary statistics, not individual-level data:

- $\hat{\beta}_j$: effect size for variant j
- $SE(\hat{\beta}_j)$: standard error
- $Z_j = \hat{\beta}_j / SE(\hat{\beta}_j)$: z-score
- $\chi_j^2 = Z_j^2$: chi-square statistic

Question: Can we infer biological mechanisms from summary statistics alone?

From univariate to multivariate effects

Underlying assumption: traits are polygenic

$$Y_i = \sum_{j=1}^p X_{ij}\theta_j + \epsilon$$

Problem: Univariate GWAS gives us $\hat{\beta}_j$ for each variant j separately

Goal: Understand the true multivariate effects θ_j from summary statistics

Challenge: LD structure confounds the relationship between $\hat{\beta}_j$ and θ_j

LD-score: Relating χ^2 statistics to heritability

What is a generative model for χ_j^2 statistics?

We can show that the univariate z-score relates to multivariate effects:

$$Z_j = \frac{\sqrt{n}}{\sigma} \sum_k R_{jk} \theta_k + \epsilon_j$$

where R is the LD matrix and $\epsilon \sim \mathcal{N}(0, 1)$.

LD-score: Relating χ^2 statistics to heritability

What is a generative model for χ_j^2 statistics?

$$\mathbb{E}[\chi_j^2] = \mathbb{E}[Z_j^2] = \mathbb{E}\left(\sqrt{n} \sum_k R_{jk} \theta_k + \epsilon_j\right)^2$$

Bulik-Sullivan *et al.*, *Nature Genetics* (2014)

LD-score: Relating χ^2 statistics to heritability

What is a generative model for χ_j^2 statistics?

$$\mathbb{E}[\chi_j^2] = \mathbb{E}[Z_j^2] = \mathbb{E}\left(\sqrt{n} \sum_k R_{jk} \theta_k + \epsilon_j\right)^2$$

If the effects are independent, i.e., $\mathbb{E}[\theta_k \theta_j] = 0$ for all $k \neq j$,

$$\mathbb{E}[\chi_j^2] = n \underbrace{\sum_k R_{jk}^2}_{\text{LD-score}} \mathbb{E}[\theta_k^2] + 1$$

Bulik-Sullivan *et al.*, *Nature Genetics* (2014)

Baseline LD-score to measure polygenic heritability

Assuming that all the variants equally contribute,

$$\mathbb{E}[\theta_k^2] = \tau/p,$$

where p is the total number of SNPs.

We get

$$\mathbb{E}[\chi_j^2] = \underbrace{n}_{\text{sample size}} \underbrace{l_j}_{\text{LD score}} \underbrace{\frac{\tau}{p}}_{\text{per SNP heritability}} + 1$$

Regression model: Regress observed χ_j^2 on LD scores l_j to estimate τ (heritability)

Bulik-Sullivan *et al.*, *Nature Genetics* (2014)

Stratified LD-score regression (S-LDSC)

Question: Which genomic annotations explain GWAS heritability?

Method: Stratified LD score regression (Finucane et al. 2015)

- Partition genome by annotations (enhancers, promoters, histone marks)
- Test which annotations are enriched for heritability
- Uses summary statistics + LD information

Tool: ldsc Python package from Broad Institute

Take-home messages

- ① Summary statistics enable powerful analyses without individual-level data
- ② LD score regression connects GWAS signals to heritability
- ③ Methods like SuSiE, coloc, and TWAS integrate multiple data types
- ④ Regression is fundamental to understanding genetic architecture

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